

Anomalous Is Not Malicious: A Provable Performance Ceiling for Unsupervised NIDS and a Two-Stage Architecture for Zero-Day Intrusion Detection

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Abstract—Unsupervised network intrusion detection systems (NIDS) consistently plateau at 55–70% recall on real-world benchmarks, a degradation widely attributed to parameter misconfiguration. We prove this ceiling is *structural*: for any threshold-based detector, the maximum achievable Youden index (TPR – FPR) equals the Kolmogorov–Smirnov (KS) distance between the class-conditional anomaly-score distributions (Theorem 1). On the CTU-13 botnet dataset, Isolation Forest attains KS = 0.445; no threshold calibration can exceed this bound. The root cause is the *anomaly–malice mismatch*: unsupervised detectors measure statistical rarity, whereas stealthy attacks are engineered to appear statistically normal. Replacing the anomaly notion — isolation by reconstruction, via an autoencoder trained exclusively on normal traffic — raises the ceiling to KS = 0.860 (F_1 : 0.617 $\rightarrow$ 0.928), and the deployed detector operates at its theoretical ceiling (balanced accuracy 0.929 vs. bound 0.930). Cross-dataset evaluation on UNSW-NB15, however, shows no single anomaly notion generalises across attack families. We therefore derive a closed-form law for OR-fusion of conditionally independent detectors (Theorem 2): miss rates multiply. The law predicts the fused recall of our dual-head detector (flow autoencoder $\cup$ temporal autoencoder) to three decimal places (predicted 0.488, measured 0.491; miss-redundancy $\rho_{\text{miss}} = 0.994$), lifting Stage-1 recall from 8.6% to 49.1% and, on hidden attack families, Backdoor from 8% to 41% and Analysis from 3% to 35%. Finally, we replace the arbitrary percentile threshold with an extreme-value-theoretic threshold (Theorem 3): a generalized Pareto tail fit ($\hat{\xi} = 1.12$, infinite variance) yields calibrated false-positive rates where percentile thresholds overshoot by $3.8\times$. The resulting two-stage architecture — fused zero-day net, gradient-boosted known-attack classifier, and a zero-day candidate queue — achieves $F_1 = 0.945$ on CTU-13 with 87% false-positive reduction. In a controlled simulation withholding four attack families from the classifier entirely, the dual-head pipeline raises end-to-end zero-day routing of Backdoor from 5.0% to 27.8% and Analysis from 2.2% to 23.8%, with zero-day queue precision of 86.7%; of all hidden-family flows flagged by Stage 1, fewer than 1% are cleared as benign — the remainder reach an analyst either through the queue or as (family-misattributed) attack alerts.

Index Terms—Network intrusion detection, anomaly detection, zero-day attacks, Kolmogorov–Smirnov distance, extreme value theory, detector fusion, Isolation Forest, autoencoder, CTU-13, UNSW-NB15.

I. INTRODUCTION

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SIGNATURE-BASED intrusion detection fails categorically against zero-day attacks: no signature exists for an attack no one has seen. Anomaly-based NIDS promise the alternative — model normal traffic, flag deviations — yet two decades of published results place unsupervised recall stubbornly in the 55–70% band [3], [5]. The standard prescriptions (tune the contamination parameter, add features, deepen the model) do not move this number. This paper explains why, proves the limitation formally, and presents a validated architecture that circumvents it.

Our contributions:

- 1) **A provable ceiling (Sec. III).** For threshold detectors, $\sup_t J(t) = \text{KS}$ exactly. Corollaries give a balanced-accuracy ceiling $(1 + \text{KS})/2$ and a recall bound at any false-alarm budget. On CTU-13 our autoencoder attains its ceiling, proving threshold optimisation is exhausted and only representation change can help.
- 2) **A fusion law (Sec. III).** Under conditional independence, OR-fused detector misses multiply. The law is verified on real traffic to three decimal places and supplies a closed-form criterion for when fusing two detectors is beneficial.
- 3) **Parameter-free thresholds (Sec. III).** A peaks-over-threshold GPD fit converts any target false-positive budget into a calibrated threshold, including budgets below the empirical resolution of the training data.
- 4) **A validated two-stage system (Sec. V).** Dual-head Stage 1 (flow + temporal autoencoders, fused per the law), a supervised Stage 2 over the flagged pool, and an explicit zero-day candidate queue, evaluated with a hold-out zero-day simulation on UNSW-NB15.

II. RELATED WORK

Isolation Forest [1] and extensions [8]; LOF [2]; autoencoder NIDS [6], [7]; EVT for anomaly thresholds [9]; CTU-13 [3] and UNSW-NB15 [4] benchmarks. Our work differs by diagnosing the ceiling formally before proposing remedies, and by validating each closed-form prediction on real traffic.

III. THEORETICAL FRAMEWORK

Let S be a scalar anomaly score (higher = more anomalous), $Y \in \{0, 1\}$ the true class (1 = attack), and $F_A(t) =$

$P(S \leq t \mid Y=1)$, $F_N(t) = P(S \leq t \mid Y=0)$ the class-conditional CDFs. A threshold detector decides “attack” iff $S > t$; its operating point is $\text{TPR}(t) = 1 - F_A(t)$, $\text{FPR}(t) = 1 - F_N(t)$, and its Youden index is $J(t) = \text{TPR}(t) - \text{FPR}(t)$.

A. The KS Ceiling

Theorem 1 (KS ceiling). *For any threshold detector on S , in either orientation,*

$$\sup_t |J(t)| = \sup_t |F_N(t) - F_A(t)| = \text{KS}(F_A, F_N),$$

and the bound is invariant under any monotone transformation of S .

Proof. $J(t) = (1 - F_A(t)) - (1 - F_N(t)) = F_N(t) - F_A(t)$; take suprema over t and over the two orientations. Monotone maps preserve both CDFs’ level sets, hence KS. $\square$

Corollary 1 (Balanced-accuracy ceiling). $\max_t \text{BA}(t) = \frac{1}{2}(1 + \text{KS})$.

Corollary 2 (Recall at a false-alarm budget). $\text{TPR}(t) \leq \text{FPR}(t) + \text{KS}$ for every t ; hence at budget $\text{FPR} \leq \alpha$, recall is at most $\alpha + \text{KS}$.

Corollary 3 (AUC bound). $\text{AUC} \leq \frac{1}{2} + \text{KS}$.

Scope. For arbitrary measurable decision regions the bound is the total variation distance $\text{TV} \geq \text{KS}$, with equality under monotone likelihood ratio; deployed detectors are threshold rules, so KS is the operative ceiling.

Empirical validation (CTU-13, held-out test). The empirical $\sup_t J(t)$ equals the two-sample KS statistic to machine precision for both detectors (difference $\leq 5.6 \times 10^{-17}$). Corollary 1: autoencoder KS = 0.863 gives ceiling 0.931; observed balanced accuracy 0.929 — *the deployed detector operates at its theoretical ceiling*. Corollary 2: Isolation Forest (KS = 0.457) cannot exceed 50.7% recall at a 5% false-alarm budget, explaining the widely reported plateau.

B. OR-Fusion of Conditionally Independent Detectors

Let $B_i = \{S_i > t_i\}$, $i = 1, 2$, be the flag events of two detectors with operating points $(\text{TPR}_i, \text{FPR}_i)$, and assume conditional independence (CI) given each class: $P(B_1 \cap B_2 \mid Y=y) = P(B_1 \mid y)P(B_2 \mid y)$.

Theorem 2 (Fusion law). *Under (CI), the OR-fused detector satisfies:*

- (a) Miss-product law: $1 - \text{TPR}_{\text{OR}} = (1 - \text{TPR}_1)(1 - \text{TPR}_2)$;
- (b) Subadditive false alarms: $\text{FPR}_{\text{OR}} = 1 - (1 - \text{FPR}_1)(1 - \text{FPR}_2) \leq \text{FPR}_1 + \text{FPR}_2$;
- (c) Fusion-benefit criterion: $J_{\text{OR}} > J_1 \iff (1 - \text{TPR}_1)\text{TPR}_2 > (1 - \text{FPR}_1)\text{FPR}_2$.

Proof. (a),(b): apply (CI) to the complements — a union miss is a simultaneous miss. (c): $J_{\text{OR}} - J_1 = (1 - \text{TPR}_1)\text{TPR}_2 - (1 - \text{FPR}_1)\text{FPR}_2$ by direct expansion. $\square$

Proposition 1 (Mixture limit). *If the attack population is a mixture in which subfamily k is invisible to the other head*

(likelihood ratio $\equiv 1$), the Neyman–Pearson optimal region degenerates to the union $B_1 \cup B_2$: OR-fusion is exactly optimal for heterogeneous attack families.

We diagnose (CI) with the miss-redundancy coefficient $\rho_{\text{miss}} = P(\bar{B}_1 \cap \bar{B}_2 \mid Y=1) / [P(\bar{B}_1 \mid Y=1)P(\bar{B}_2 \mid Y=1)]$ ($\rho = 1$: independent blind spots) and the conditional mutual information $I(S_1; S_2 \mid Y)$ estimated by within-class quantile binning.

Empirical validation (UNSW-NB15, flow-AE head vs. temporal-AE head). Head operating points $\text{TPR}_1 = 0.086$, $\text{TPR}_2 = 0.440$ (both at 95th-percentile thresholds). The law predicts union recall $1 - (1 - 0.086)(1 - 0.440) = 0.488$; measured: 0.491. Overall $\rho_{\text{miss}} = 0.994$ and $I(S_1; S_2 \mid Y) = 0.28$ bits — the two anomaly notions are empirically conditionally independent. Per-family $\rho_{\text{miss}} \in [1.00, 1.13]$. The fusion criterion holds with a $6\times$ margin. Fused Stage-1 recall on hidden families: Backdoor 8% $\rightarrow$ 41%, Analysis 3% $\rightarrow$ 35%.

C. Calibrated Thresholds via Extreme Value Theory

Theorem 3 (Pickands–Balkema–de Haan, applied). *If the normal-score distribution lies in the maximum domain of attraction of an extreme value law, the excess distribution over a high threshold u converges uniformly to a generalized Pareto distribution $G_{\xi, \sigma}(x) = 1 - (1 + \xi x / \sigma)^{-1/\xi}$. Consequently, with $p_u = P(S_N > u)$ estimated empirically and $(\hat{\xi}, \hat{\sigma})$ fitted by maximum likelihood on the excesses, the threshold achieving a target false-positive rate q is*

$$t_q = u + \frac{\hat{\sigma}}{\hat{\xi}} \left[(p_u/q)^{\hat{\xi}} - 1 \right] \quad (\hat{\xi} \neq 0).$$

Empirical validation (UNSW-NB15 flow-AE, $n = 56,000$ training normals, $u = 90$ th percentile). The fitted shape is $\hat{\xi} = 1.12$: the reconstruction-error tail is so heavy its variance is infinite, which is precisely why percentile thresholds miscalibrate. At target $q = 10^{-3}$ the GPD threshold realises $\text{FPR} = 4.9 \times 10^{-4}$ on held-out normals while the percentile threshold realises 3.8×10^{-3} ($3.8\times$ over budget). The GPD extrapolates analytically to budgets below the $1/n$ resolution of the data, removing the last free parameter from Stage 1.

IV. DATASETS AND EXPERIMENTAL PROTOCOL

CTU-13 [3]: 38,898 botnet flows, 53,314 normal, 57 CFlowMeter features; balanced 6k/6k samples, 70/30 stratified split. UNSW-NB15 [4]: 175,341 train / 82,332 test flows, nine attack families, 39 Argus features. All preprocessing (median imputation, $\log(1+x)$ on skewed columns, standardisation) fitted on training partitions only.

V. THE TWO-STAGE ARCHITECTURE

Stage 1: dual autoencoder heads (flow features; temporal/behavioural features), each trained on normal traffic only, fused by OR per Theorem 2. Stage 2: gradient-boosted classifier over the flagged pool, trained on the five *known* attack families only; flows below confidence $\tau = 0.55$ enter the *zero-day candidate queue*. Four families (Backdoor, Analysis, Shellcode, Worms) are withheld from Stage 2 at all times.

TABLE I
STAGE-1 CONFIGURATIONS. UNION IS THE PROPOSED HEAD;
INTERSECTION IS THE CONTROL.

Config	TPR	FPR	Backdoor	Analysis
Flow head only	0.086	0.086	0.08	0.03
Temporal head only	0.440	0.073	0.39	0.35
Union (ours)	0.491	0.143	0.41	0.35
Intersection (control)	0.034	0.016	0.06	0.02

TABLE II
FULL PIPELINE, IDENTICAL STAGE 2, ONLY STAGE 1 CHANGES.

Metric	Flow-only	Dual-head
Pipeline recall	0.084	0.488
Pipeline precision	0.850	0.939
Pipeline F_1	0.152	0.643
Zero-day queue precision	0.753	0.867
Backdoor end-to-end routing	5.0%	27.8%
Analysis end-to-end routing	2.2%	23.8%
Shellcode end-to-end routing	0.5%	2.1%

A. Stage-1 Ablation (UNSW-NB15 test)

Table I confirms Theorem 2: the union recall (0.491) matches the miss-product prediction (0.488), and the intersection control collapses as expected.

B. End-to-End Zero-Day Simulation

The dual head raises pipeline F_1 by $4.2\times$ while increasing precision (Table II). Of the 518 hidden-family flows flagged by Stage 1, 332 route to the zero-day queue, 182 are alerted as family-misattributed known attacks, and only 4 ($< 1\%$) are cleared as benign: hidden attacks are misattributed, not missed. Family misattribution is an honest limitation — the alert fires, but the analyst receives the wrong attack label. Shellcode remains largely undetected at Stage 1 (2.1%): a single-shot exploit leaves neither a flow-statistical nor a temporal footprint, delimiting what flow-level detection can achieve.

On CTU-13 the single-head pipeline attains $F_1 = 0.945$ with an 87% false-positive reduction over Stage 1 alone.

VI. DISCUSSION AND LIMITATIONS

VII. CONCLUSION

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